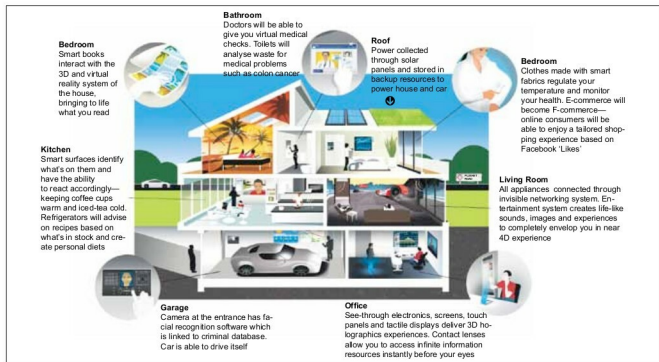


Ubiquitous Computing

Getting Closer To A Distant Dream

Ubiquitous computing is a subject we revisit every year because we are as curious as you to see how the dream is shaping up! Today, we find that the building blocks are all there as also a few successful demonstrations, but a lot of disparate issues need to be resolved before the jigsaw falls into place



Ubiquitous computing environment of the future

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Those who have not heard of ubiquitous computing, or ubi-comp, often ask whether it is a new technology. Indeed, it is not merely a technology but a new paradigm of computing that Mark Weiser of Xerox PARC conceptualised about two-and-a-half decades ago. Prof. Ken Sakamura is also known to have proposed and demonstrated the concept in Japan through the TRON Smart House model in the 1980s.

Some rightly call ubiquitous com-

puting the post-desktop paradigm of computing, while others call it the Internet of Things (IoT). In simple terms, ubi-comp represents a situation wherein technology is everywhere, but in an unobtrusive way.

In a ubi-com world, there will be innumerable computers embedded everywhere, in everyday devices, from traffic signals and air-conditioners to coffee mugs and even clothes. There will be wearable centimetre-sized devices or tags, hand-held decimetre-sized devices and metre-sized interactive display devices. There will be micro-electro-mechanical systems from nanometre

to millimetre scales and organic fabric-like devices. Dust-like devices will do some computing or communications without user interfaces. What is more, all these devices will have a unique ID or address and will connect through an intelligent network. The whole system will be so smart that an air-conditioner will be able to adjust the temperature based on the feedback from a sensor embedded in your dress.

This vision does not seem as incredible today as it seemed a decade ago, due to the technological developments that have happened in the recent past. For one, the number of connected